CIVIL ENGINEERING · KUWAIT UNIVERSITY · EXAM PREP

Soil Mechanics

Chapters 1–5, made visual, animated and easy to remember.

Exam: 9 November

Based on Das & Sobhan, 9th ed.

GRAVEL · > 4.75 mm

SAND · 4.75 – 0.075 mm

SILT · 0.075 – 0.002 mm

CLAY · < 0.002 mm

THE MAP

Five chapters, one story

Soil starts as rock, gets measured, gets classified. Each chapter builds on the one before.

1

History

Who found what, and when

2

Origin and grain size

Sieve, hydrometer, Cu and Cc

3

Weight and volume

Phase diagram: e, n, S, w, γ

4

Plasticity

LL, PL, SL, PI, LI and structure

5

Classification

USDA, AASHTO and USCS

Each chapter ends with a quick check. Answer out loud before you click.

CHAPTER 1

The story of soil mechanics

1

From pyramids to Terzaghi: how guessing turned into a science.

4 periods | Who did what | 1925

CHAPTER 1 · DEFINITIONS

Four words you must define

Soil

Uncemented grains of mineral and decayed organic matter, with liquid and gas in the gaps between them.

Soil mechanics

The science of soil's physical properties and how a mass of soil behaves under forces.

Soils engineering

Putting soil-mechanics principles to work on practical problems.

Geotechnical engineering

Soil mechanics plus rock mechanics, applied to foundations, retaining structures and earth structures.

Soil is both a construction material and a foundation material.

CHAPTER 1 · TIMELINE

Four periods, 1700 to 1927

1700–1776

Preclassical

Natural slopes, unit weights and rule-of-thumb earth pressure.

Gautier · Belidor · Gadroy

1776–1856

Classical I

Rigorous earth-pressure theory, mostly French.

Coulomb · Poncelet · Collin

1856–1910

Classical II

Lab tests on sand: permeability, stress distribution, dilatancy.

Darcy · Boussinesq · Reynolds

1910–1927

Modern

The behaviour of clays is finally understood.

Atterberg · Fellenius · Terzaghi

Memory hook: guesswork → Coulomb’s maths → sand in the lab → clay.

CHAPTER 1 · WHO DID WHAT

Eight names to know cold

Coulomb
1776
Sliding wedge with friction and cohesion

Rankine
1857
Simplified Coulomb's earth-pressure theory

Darcy
1856
Permeability law from sand-filter tests

Boussinesq
1885
Stress under a load in an elastic half-space

Atterberg
about 1908–1911
Clay fraction (< 2 μm); liquid, plastic, shrinkage limits

Fellenius
1918 and 1926
Circular slip surfaces; φ = 0 for saturated clay slopes

Terzaghi
1925
Consolidation theory; book Erdbaumechanik

Skempton
1948 and 1954
φ = 0 concept; pore-pressure parameters A and B

Also: Poncelet (1840, the symbol φ), Collin (1846, clay slips), Reynolds (1887, dilatancy)

CHAPTER 1 · A FAMOUS FOUNDATION PROBLEM

Why the Tower of Pisa leans

- Begun in **1173**, built in stages over 200+ years.
- About **15,700 t** on a base roughly 20 m wide.
- A weak **clay layer about 11 m deep** compressed unevenly.
- Over 5 m out of plumb on 54 m: about **5.5°**.
- Fixed by removing about **70 t of soil** under the **north** side: tilt now about 5°.



Bologna: Garisenda 48 m, tilt about 4°. Asinelli 97 m, about 1.3°.

CHAPTER 1 · QUICK CHECK

Answer first, then click

Father of modern soil mechanics, and the year of his book?	Terzaghi, 1925
Which period was about lab tests on sand?	Classical II, 1856–1910
Name the three Atterberg limits.	Liquid, plastic, shrinkage
Why does the Tower of Pisa lean?	Weak clay about 11 m deep

CHAPTER 2

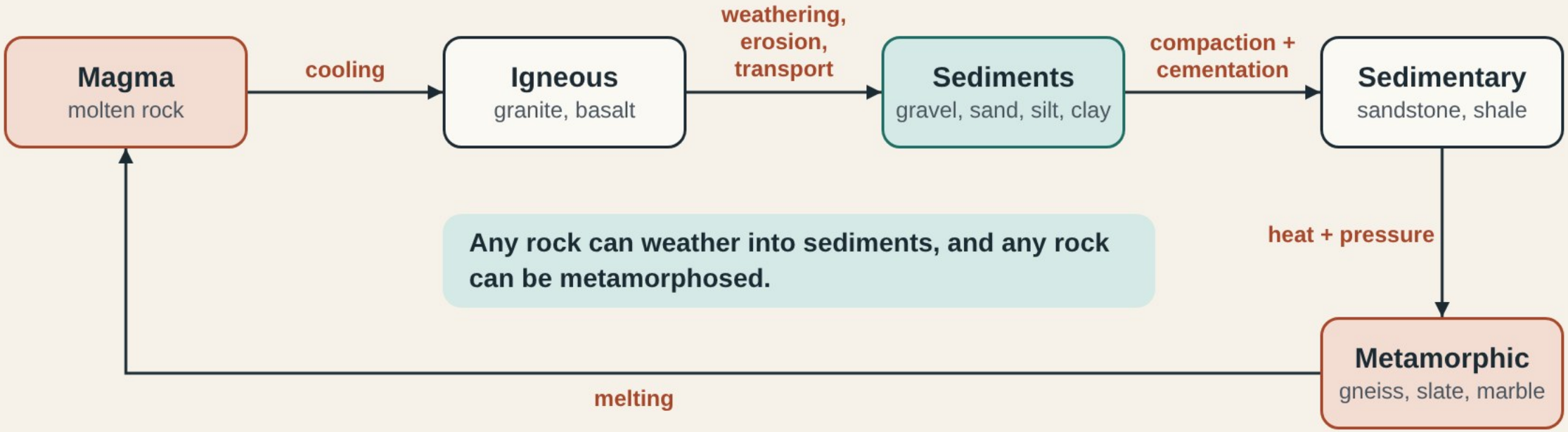
Origin of soil and grain size

2

Rocks break down, clays form, and we measure the pieces.

Rock cycle | Clay minerals | Sieve and hydrometer

Every soil starts as a rock



CHAPTER 2 · WEATHERING

Break it or change it

Mechanical (physical)

Rock is broken into smaller pieces. Its composition does not change.

Agents: temperature swings, freeze–thaw, glaciers, wind, running water, waves.

Product: gravel, sand and silt.

Chemical

Minerals react and turn into new minerals.

Agents: water + CO_2 (carbonic acid), dissolved salts, organic acids.

Product: clay minerals, e.g. orthoclase → kaolinite.

Bowen's series: olivine → augite → hornblende → biotite … orthoclase → muscovite → quartz. Top of the series weathers first; quartz is the most resistant.

Transported soils: glacial, alluvial (rivers), lacustrine (lakes), marine, aeolian (wind), colluvial (gravity). Residual soils stay where they formed.

Three clays, three personalities

Kaolinite

1:1 layers, about 7.2 Å

Strong hydrogen bonds

Stable, low swelling

Illite

2:1 layers, about 10 Å

Potassium ion bonds

Intermediate behaviour

Montmorillonite

2:1 layers, variable spacing

Weak bonds: water enters

Very high swelling

Specific surface (surface area per gram)



Where does sand end and silt begin?

System (mm)	Gravel	Sand	Silt	Clay
MIT	> 2	2 – 0.06	0.06 – 0.002	< 0.002
USDA	76.2 – 2	2 – 0.05	0.05 – 0.002	< 0.002
AASHTO	76.2 – 2	2 – 0.075	0.075 – 0.002	< 0.002
Unified (USCS)	76.2 – 4.75	4.75 – 0.075	fines: < 0.075	(silt + clay)

Sieve No. 4 = 4.75 mm · No. 10 = 2 mm · No. 40 = 0.425 mm · No. 200 = 0.075 mm

Clay-size is not the same as clay mineral: some very fine quartz is also clay-sized (< 2 μm).

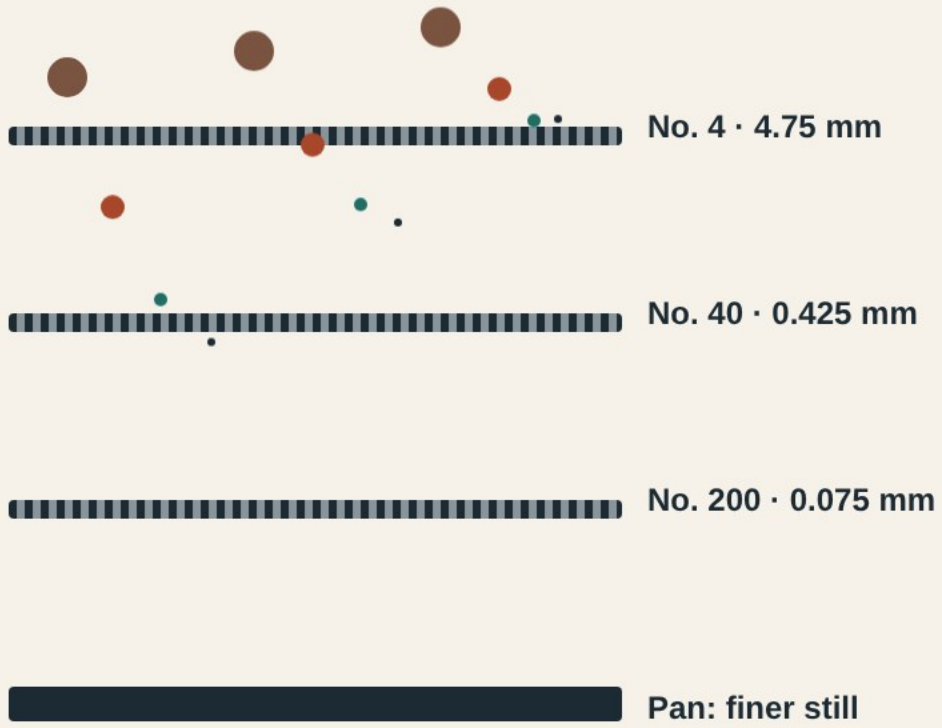
CHAPTER 2 · SIEVE ANALYSIS

Shake, weigh, repeat

1. Oven-dry the soil. Stack sieves: coarsest on top, pan at the bottom.
2. Shake, then weigh what stays on each sieve.
3. Add up the mass retained so far, then subtract from the total.

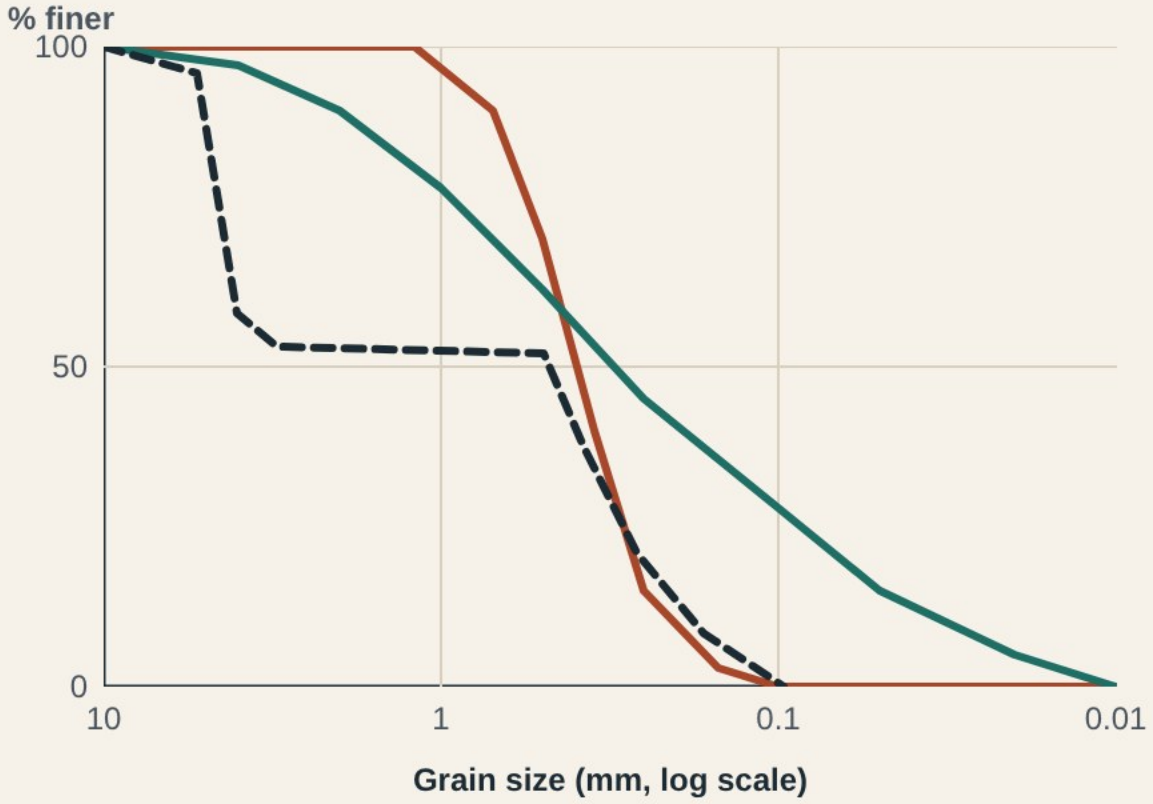
$$\% \text{ finer} = (\text{total} - \text{cumulative retained}) \div \text{total} \times 100$$

Plot % finer on a normal axis against grain size on a log axis. Sieves work above 0.075 mm (No. 200).



CHAPTER 2 · GRAIN-SIZE CURVE

Read the shape of the curve



Poorly (uniformly) graded: steep curve, grains almost all one size.

Well graded: gentle curve, wide range of sizes. Needs Cu > 4 (gravel) or > 6 (sand), and 1 ≤ Cc ≤ 3.

Gap graded (dashed): a flat part means a size range is missing.

$$Cu = D60 / D10$$

$$Cc = D30^2 / (D60 \times D10)$$

CHAPTER 2 · WORKED EXAMPLE

450 g of soil: find Cu and Cc

Sieve	mm	Retained (g)	% finer
4	4.75	0	100
10	2.00	21.6	95.2
20	0.850	49.5	84.2
40	0.425	102.6	61.4
60	0.250	89.1	41.6
100	0.150	95.6	20.4
200	0.075	60.4	6.9
Pan	–	31.2	0

1. READ THE CURVE

D60 = 0.41, D30 = 0.185, D10 = 0.09 mm

2. UNIFORMITY

$Cu = 0.41 / 0.09 = 4.56$

3. CURVATURE

$Cc = 0.185^2 / (0.41 \times 0.09) = 0.93$

Cu < 6 and Cc < 1, so this sand is poorly graded.

CHAPTER 2 · HYDROMETER ANALYSIS

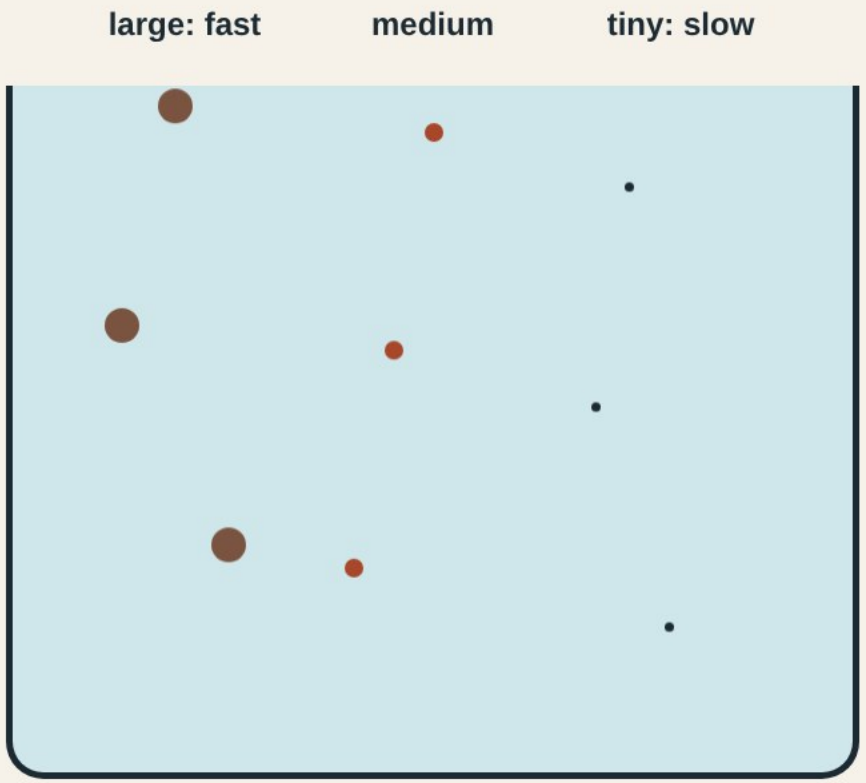
Slow settling tells us size

For grains finer than 0.075 mm, we time how fast they sink in water.

Stokes: $v = (\rho s - \rho w)\, D^2 / (18\, \eta)$

$D\ (mm) = K \sqrt{(\ L / t\ )}$ L in cm, t in min

Speed grows with D^2, so bigger grains sink first. Set-up: 50 g soil, dispersing agent (sodium hexametaphosphate), made up to 1000 mL, ASTM 152H hydrometer.



CHAPTER 2 · QUICK CHECK

Answer first, then click

Name the three rock types.	Igneous, sedimentary, metamorphic
Which clay mineral swells the most?	Montmorillonite
A sand has Cu = 3 and Cc = 1.2. Well graded?	No. Cu must exceed 6 for a sand.
What does the No. 200 sieve separate?	Sand from fines, at 0.075 mm

CHAPTER 3

Weight–volume relationships

3

Soil is solids, water and air. Draw it, then solve it.

Phase diagram | S·e = w·Gs | Relative density

Pull the soil apart into three layers



Always draw this first: volumes on the left, weights on the right. The weight of air is neglected.

CHAPTER 3 · DEFINITIONS

Six ratios that run the chapter

e = Vv / Vs
Void ratio. Can be above 1 (soft organic clays).

w = Ww / Ws
Water content. Divide by the weight of SOLIDS.

n = Vv / V
Porosity. Always below 1 (below 100%).

γ = W / V
Moist (total) unit weight = Ws (1 + w) / V.

S = Vw / Vv
Degree of saturation. 0% is dry, 100% is saturated.

γd = Ws / V
Dry unit weight, and γd = γ / (1 + w).

e = n / (1 − n) n = e / (1 + e) γw = 9.81 kN/m³

One equation links everything

$$S \cdot e = w \cdot G_s$$

Where it comes from: take Vs = 1

STEP 1
Vs = 1, so Vv = e and V = 1 + e

STEP 2
Ws = Gs · γw

STEP 3
Vw = S · e, so Ww = S · e · γw

STEP 4
w = Ww / Ws = S·e / Gs, so S·e = w·Gs

DRY
$\gamma_d = G_s \gamma_w / (1 + e)$

MOIST
$\gamma = (1 + w) G_s \gamma_w / (1 + e)$

SATURATED
$\gamma_{sat} = (G_s + e) \gamma_w / (1 + e)$

Saturated (S = 1): e = w·Gs. Dry (S = 0): w = 0.

CHAPTER 3 · PROBLEM RECIPE

Five steps for any problem

1

Sketch

Draw air, water and solids. Volumes left, weights right.

2

Pick a basis

Vs = 1 if you know e. V = 1 if you know n. Or use the given volume.

3

Fill in

Use Gs, w and γw to write every box you can.

4

Link

Use S·e = w·Gs to tie saturation to the rest.

5

Check units

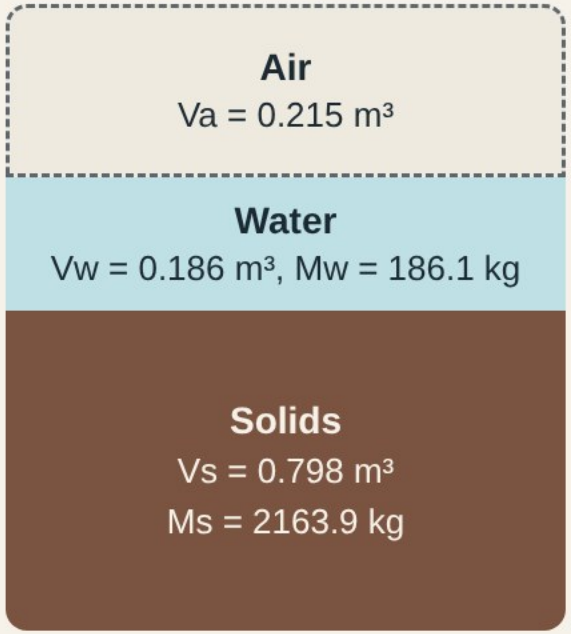
γw = 9.81 kN/m³ (62.4 lb/ft³). ρw = 1000 kg/m³.

Typical Gs: quartz sand about 2.65, clays and silts 2.6 to 2.9.

Density to unit weight: γ (kN/m³) = 9.81 × ρ (kg/m³) ÷ 1000.

CHAPTER 3 · WORKED EXAMPLE

Fill in the phase diagram



STEP 1 · MASS OF SOLIDS

Ms = M / (1 + w) = 2350 / 1.086 = 2163.9 kg

STEP 2 · VOLUME OF SOLIDS

Vs = Ms / (Gs ρw) = 2163.9 / 2710 = 0.798 m³

STEP 3 · VOIDS

Vv = 1.2 − 0.798 = 0.402 m³, e = 0.503, n = 0.335

STEP 4 · WATER AND SATURATION

Mw = 186.1 kg, Vw = 0.186 m³, S = 0.186 / 0.402 = 46.3%

Given: V = 1.2 m³, M = 2350 kg, w = 8.6%, Gs = 2.71. Find e, n, S.

How dense is this sand, really?

$$Dr = (emax - e) / (emax - emin)$$

Same thing with yd: $Dr = [(yd - yd,min) / (yd,max - yd,min)] \times [yd,max / yd]$

Very loose	Loose	Medium	Dense	Very dense
0–15%	15–50%	50–70%	70–85%	85–100%

FIELD EXAMPLE (3.7)

emax 0.75, emin 0.40, e = 0.67: Dr = 22.9%, loose

CORRELATION EXAMPLE (3.8)

emax 0.875, emin 0.525, e = 0.66: Dr = 61.4%, medium

emax = loosest state = lowest yd. emin = densest state = highest yd. Granular soils only.

CHAPTER 3 · QUICK CHECK

Answer first, then click

Saturated soil, Gs = 2.70, w = 20%. Void ratio e?	e = w·Gs = 0.54
Porosity n = 0.40. Void ratio e?	e = n/(1−n) = 0.667
Which is larger for a moist soil, γ or γd?	γ, because γd = γ/(1+w)
What is Dr when e = emax?	0%, the loosest state

CHAPTER 4

Plasticity and soil structure

4

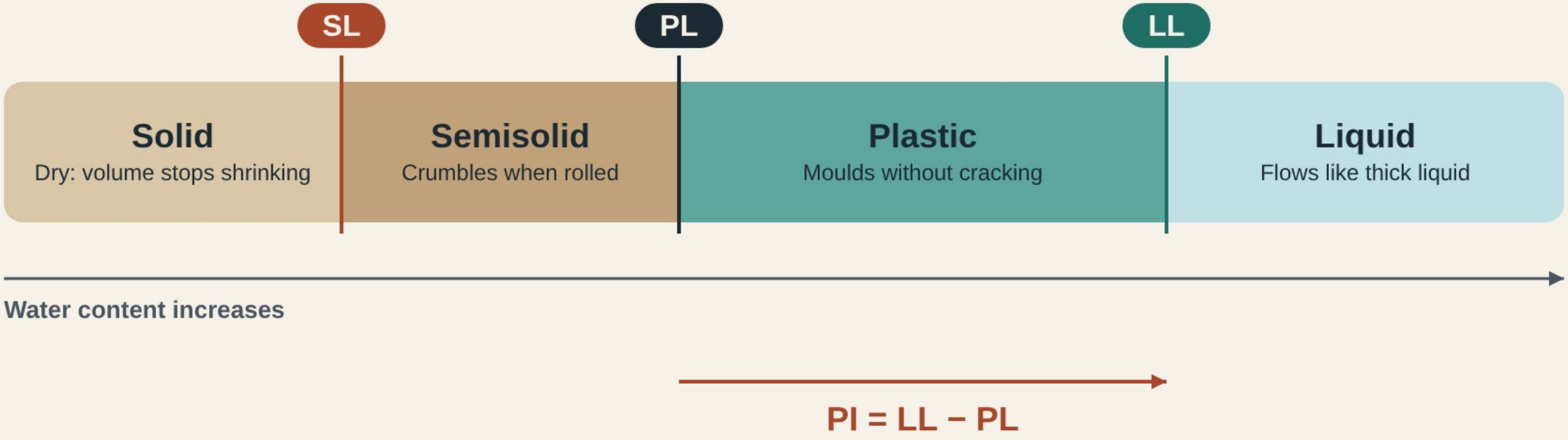
How clay changes as it gets wetter, and how its grains are arranged.

LL · PL · SL

PI · LI · activity

Plasticity chart

Add water, watch the clay change



SL: solid ↔ semisolid. PL: semisolid ↔ plastic. LL: plastic ↔ liquid. All are water contents in %.

Two ways to find the LL

Casagrande cup, top view of the groove



Blows: 13

Tap, tap, tap…

Casagrande cup (ASTM D-4318)

Cup drops 10 mm. LL = water content when the groove closes 12.5 mm after 25 blows. Run 3 or more tests with N from 15 to 35, then plot w against log N (the flow curve).

Fall cone (BS 1377)

30° cone, 80 g (0.78 N). LL = water content at 20 mm penetration in 5 s. Run 4 or more tests and plot w against the penetration d.

One point: LL = wN (N / 25)^0.121

Flow index: IF = (w1 − w2) / log(N2 / N1)

Where plasticity starts and how big it is

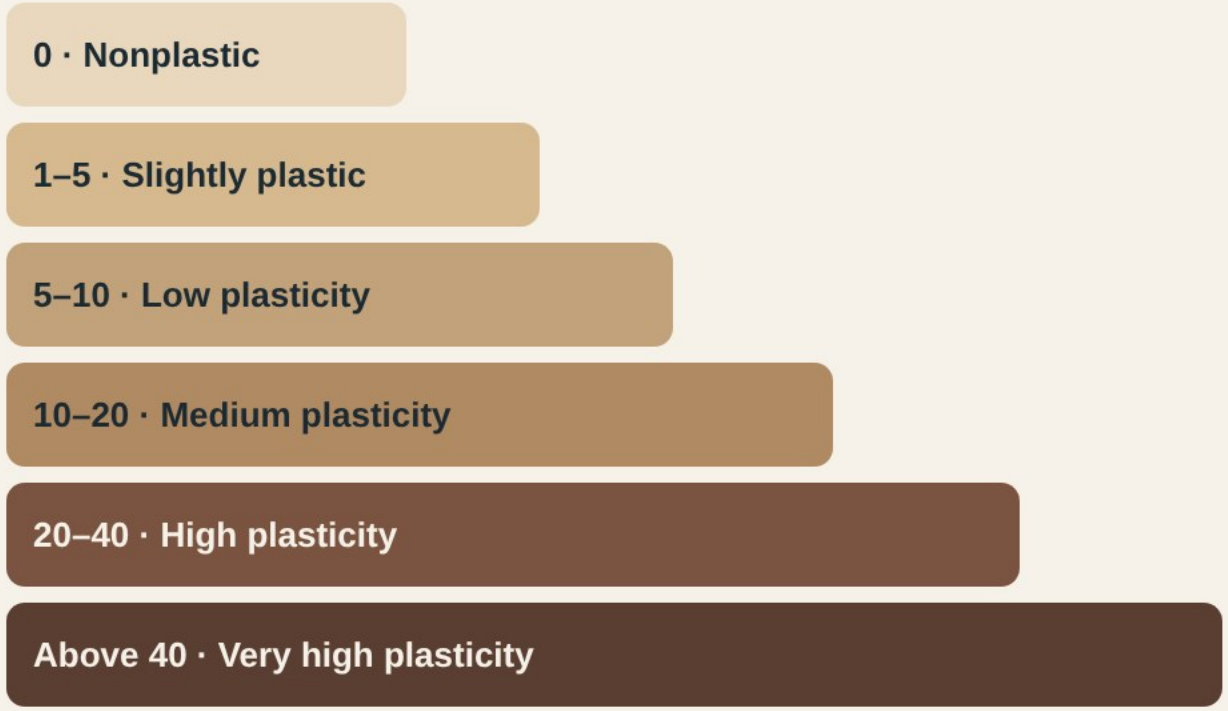
PLASTIC LIMIT (PL)

3.2 mm

Roll the soil into a thread by hand on a glass plate. PL is the water content when the thread crumbles at 3.2 mm (1/8 in.).

PI = LL − PL

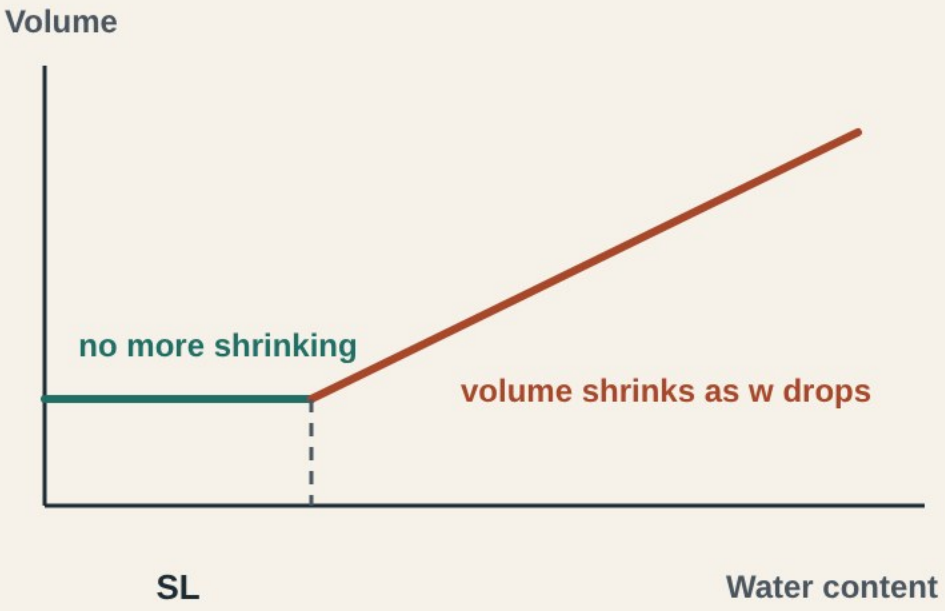
The width of the plastic range. It drives the plasticity chart.



Burmister scale, PI in %

CHAPTER 4 · SHRINKAGE LIMIT

When drying stops shrinking the soil



$$SL = wi - \Delta w$$
start water content minus water lost while shrinking

$$wi = (M1 - M2) / M2 \times 100$$
M1 = wet mass of the pat, M2 = dry mass

$$\Delta w = (Vi - Vf)\, \rho w / M2 \times 100$$
Vi = initial volume, Vf = dry volume

$$SR = M2 / (Vf\, \rho w)\ Gs = 1 / (1/SR - SL/100)$$
shrinkage ratio, then specific gravity of solids

Example 4.5: SL = 46.18 − 28.9 = 17.28%. Example 4.6: SR = 30.1 / 15.9 = 1.89 and Gs ≈ 2.81. Typical SL: montmorillonite 8.5–15, illite 15–17, kaolinite 25–29.

Where does the natural water sit?

$$LI = (w - PL) / (LL - PL)$$

$$CI = (LL - w) / (LL - PL)$$

LI < 0	0 ≤ LI ≤ 1	LI > 1
heavily overconsolidated	in the plastic range: ordinary clay	sensitive: flows when remolded

PL LL

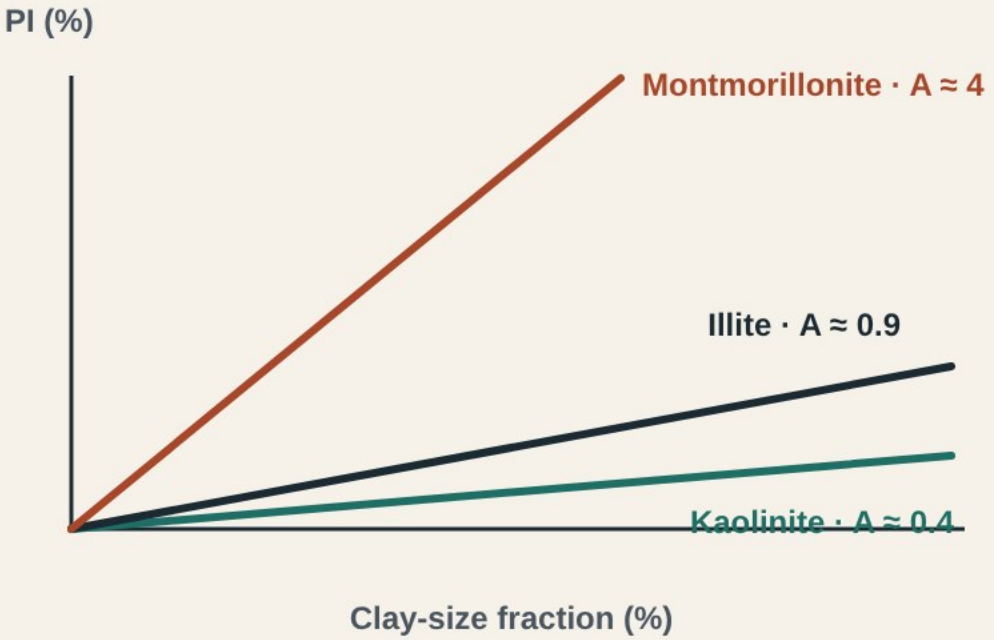
CI and unconfined compression strength (kN/m²)

CI < 0.5	0.5–0.75	0.75–1.0	1.0–1.5	CI > 1.5
< 25	25–80	80–150	150–400	> 400

LI + CI = 1. Stiffer clay means higher CI and higher strength.

CHAPTER 4 · ACTIVITY

How lively is the clay?



A = PI / (% clay-size fraction)

Clay-size = finer than 2 μm by weight. If the line misses the origin: A = PI / (% clay − C′), with C′ = 9 in Seed's tests.

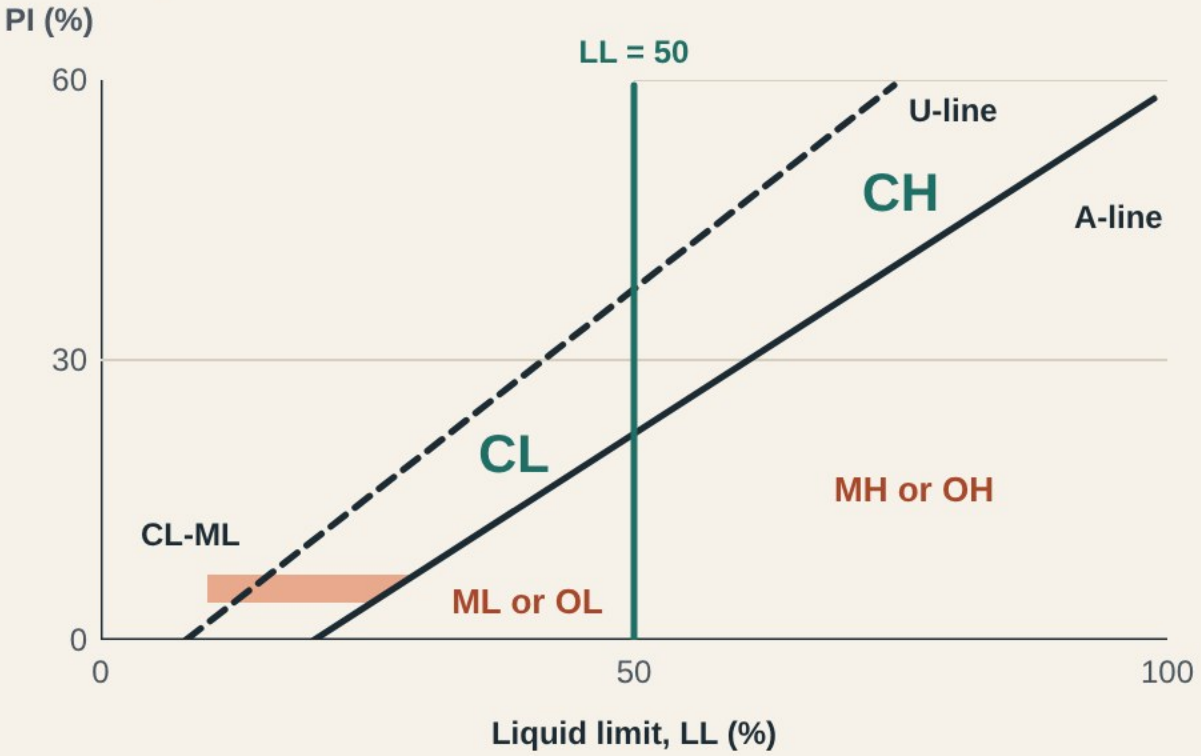
Kaolinite 0.3 – 0.5 (low swelling)

Illite 0.5 – 1.2

Montmorillonite 1.5 – 7.0 (high swelling)

Use: activity is an index of the swelling potential of a clay soil.

Clay above the line, silt below



A-line: PI = 0.73 (LL − 20)
above = clays, below = silts and organics

U-line: PI = 0.9 (LL − 8)
upper limit for any known soil

LL = 50
low plasticity (L) on the left, high (H) on the right

Shaded: 4 ≤ PI ≤ 7
just above the A-line = CL-ML (dual symbol)

CHAPTER 4 · SOIL STRUCTURE

How the grains are arranged

Sands: packing equal spheres (void ratio e)

Simple cubical · e = 0.91

Simple stagger · 0.65

Double stagger · 0.43

Pyramidal · 0.35

Honeycombed

Fine sand and silt in small arches. Big voids; collapses under heavy or shock loads.

Clays: forces between particles

Dispersed

Particles roughly parallel. Repulsion wins.

Flocculent

Edge-to-face contact, or salt cutting repulsion. Open, high void ratio. Marine clays are highly flocculent.

Domains → clusters → peds

seen with: electron microscope → light microscope → naked eye

CHAPTER 4 · QUICK CHECK

Answer first, then click

LL = 60% and PL = 25%. What is PI, and how plastic?	PI = 35, high plasticity
w = 30%, PL = 20%, LL = 40%. Find LI.	LI = 10 / 20 = 0.5
LL = 50, PI = 20. Above or below the A-line?	Below: A-line gives 21.9
Which clay mineral is the most active?	Montmorillonite, A up to 7

CHAPTER 5

Classification of soil

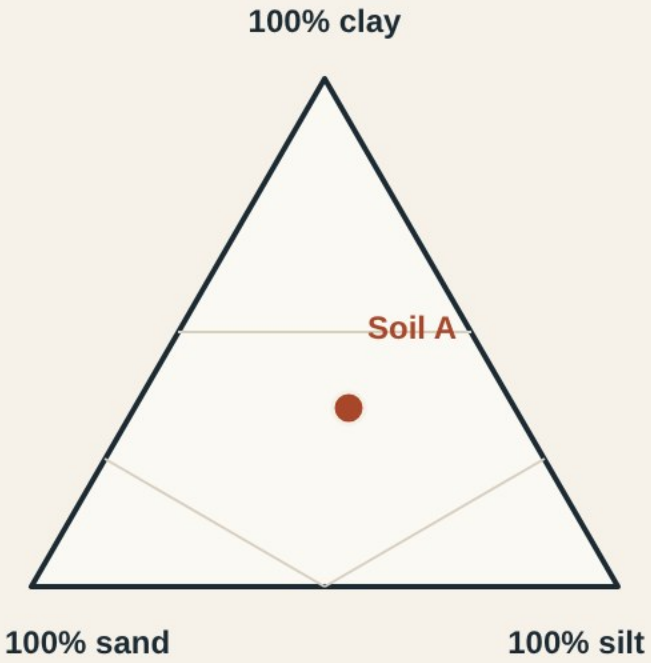
5

Give every soil a name that engineers everywhere agree on.

USDA AASHTO USCS

CHAPTER 5 · USDA TEXTURAL

Sand, silt, clay: that is all it sees



1. Gravel? Re-scale first

Modified % = (% of that fraction ÷ (100 − % gravel)) × 100

2. Plot the three percentages

Sand 2–0.05 mm, silt 0.05–0.002 mm, clay below 0.002 mm. They sum to 100.

3. Read the zone

Clay loam, sandy loam, silty clay … Add “gravelly” when gravel is significant.

Example 5.1, soil A

Gravel 12, sand 25, silt 32, clay 31 → 28.4 / 36.4 / 35.2 → gravelly clay loam

CHAPTER 5 · AASHTO

Highway subgrades in seven groups

F200 ≤ 35%
granular: A-1, A-2, A-3

F200 > 35%
silt–clay: A-4 to A-7

	LL ≤ 40	LL ≥ 41
PI ≤ 10	**A-4** silty soils	**A-5** silty soils
PI ≥ 11	**A-6** clayey soils	**A-7** clayey: -5 or -6

The same 2×2 pattern gives A-2-4, A-2-5, A-2-6, A-2-7 for granular soils with fines.

Gravel passes 75 mm and stays on No. 10. Sand passes No. 10 and stays on No. 200. Fines pass No. 200.

Run LL and PI on the fraction passing No. 40.

Go left to right through the table. The first group the soil fits is the answer.

A-7-5 if PI ≤ LL − 30. A-7-6 if PI > LL − 30.

The group index: a subgrade score

$$GI = (F200 - 35)[0.2 + 0.005\,(LL - 40)] + 0.01\,(F200 - 15)(PI - 10)$$

Higher GI means a worse subgrade. Write it in brackets: A-6(8).

Negative result → GI = 0.

Round to the nearest whole number (3.4 → 3, 3.5 → 4).

GI = 0 for A-1-a, A-1-b, A-2-4, A-2-5, A-3.

A-2-6 and A-2-7: use only 0.01 (F200 − 15)(PI − 10).

Example 5.3
F200 = 95, LL = 60, PI = 40 → A-7-6. GI = 18 + 24 = 42, so A-7-6(42).

Example 5.5
F200 = 58, LL = 30, PI = 10 → A-4. GI = 23 × 0.15 = 3.45 → 3. The book prints 4: ask which your course uses.

CHAPTER 5 · USCS COARSE-GRAINED

Sands and gravels: count the fines

First letter: G gravel · S sand · M silt · C clay · O organic · Pt peat. Second letter: W well graded · P poorly graded · L low plasticity (LL < 50) · H high plasticity (LL ≥ 50)

Under 5% fines

CLEAN: look at the curve

GW or SW needs Cu ≥ 4 (gravel) or ≥ 6 (sand) and 1 ≤ Cc ≤ 3.

Otherwise GP or SP.

5% to 12% fines

BORDERLINE: dual symbol

Gradation first, then the fines type.

Examples: SP-SC, GW-GM.

Over 12% fines

DIRTY: look at the fines

GM or SM: PI < 4, or plots below the A-line.

GC or SC: PI > 7 and on or above the A-line.

Coarse means more than 50% is retained on No. 200. Gravel if more than half of the coarse part is retained on No. 4, otherwise sand.

Group name: add “with sand” (for gravels) or “with gravel” (for sands) when the other coarse part is 15% or more.

CHAPTER 5 · USCS FINE-GRAINED

Fine soils: plot LL and PI

	On or above A-line: clay	Below A-line: silt	Organic
LL < 50 (low)	**CL** lean clay (PI > 7)	**ML** silt; CL-ML if PI 4–7 in the hatched zone	**OL** oven-dried LL / LL < 0.75
LL ≥ 50 (high)	**CH** fat clay	**MH** elastic silt	**OH** same oven-drying test

Group name: look at the % retained on No. 200

Under 15%
Base name only: lean clay.

15% to 29%
Add “with sand” or “with gravel”.

30% or more
Prefix “sandy” or “gravelly”, whichever is larger.

CHAPTER 5 · SIDE BY SIDE

AASHTO versus USCS

Feature	AASHTO	USCS
Fine-grained when	> 35% passes No. 200	≥ 50% passes No. 200
Gravel / sand split	No. 10 (2 mm)	No. 4 (4.75 mm)
Organic soils	no place for them	OL, OH, Pt
Labels	A-1 to A-7 groups	symbols like GW, SM, CH
Extra	group index (subgrade)	group name

Used by: AASHTO for highway agencies, USCS for geotechnical engineers.

Rough matches: A-1-a ↔ GW, GP · A-3 ↔ SP · A-6 ↔ CL · A-7-6 ↔ CH or CL

CHAPTER 5 · WORKED EXAMPLE

One soil, two names

58% passes No. 200, LL = 30, PI = 10, no gravel.

AASHTO

1. 58% > 35% passes No. 200, so silt–clay.
2. LL = 30 ≤ 40 and PI = 10 ≤ 10, so A-4.
3. GI = (58 − 35)[0.2 + 0.005 (30 − 40)] + 0.01 (58 − 15)(10 − 10) = 23 × 0.15 = 3.45.
4. Round to the nearest whole number: 3 (the book prints 4).

A-4(3)

USCS

1. 58% ≥ 50% passes No. 200, so fine-grained.
2. LL = 30 < 50. A-line at LL 30 is 0.73 × 10 = 7.3. PI = 10 is above it, so CL.
3. Retained on No. 200 is 42%, which is 30% or more, and sand (42%) beats gravel (0%).
4. Prefix “sandy”.

CL: sandy lean clay

CHAPTER 5 · QUICK CHECK

Answer first, then click

Which system splits gravel and sand at 2 mm?	AASHTO (USCS uses 4.75 mm)
Sand, 3% fines, Cu = 8, Cc = 1.5. USCS symbol?	SW: clean, Cu ≥ 6, Cc 1 to 3
Fine soil with LL = 65, PI = 20. USCS symbol?	MH: A-line is 32.9, PI is below
Write the A-line equation.	PI = 0.73 (LL − 20)

FORMULA RECAP

Everything on one screen

Ch. 2 · Grain size

$C_u = D_{60} / D_{10}$ $C_c = D_{30}^2 / (D_{60} D_{10})$

% finer = (total − cumulative retained) / total × 100

$D\ (mm) = K \sqrt{(L / t)}$

Ch. 3 · Weight–volume

$e = n / (1 - n)$ $S \cdot e = w \cdot G_s$

$\gamma_d = G_s \gamma_w / (1 + e) = \gamma / (1 + w)$

$D_r = (e_{max} - e) / (e_{max} - e_{min})$

Ch. 4 · Plasticity

$PI = LL - PL$ $LI = (w - PL) / (LL - PL)$

A = PI / % clay A-line: $PI = 0.73\ (LL - 20)$

$SL = w_i - \Delta w$ $SR = M_2 / (V_f \rho_w)$

Ch. 5 · Classification

$GI = (F_{200} - 35)[0.2 + 0.005\ (LL - 40)] + 0.01\ (F_{200} - 15)(PI - 10)$

USCS 50% and No. 4 · AASHTO 35% and No. 10

Fines: clean < 5%, dual 5–12%, dirty > 12%

Chapter 1 is names and dates: see the who-did-what slide.

EXAM TRAPS

Six mistakes that cost marks

WATER CONTENT

Dividing by total weight.

Divide by the weight of solids: w = Ww / Ws.

GRADATION

Checking only Cu.

Check Cu and Cc (1 to 3). Gravel Cu ≥ 4, sand Cu ≥ 6.

UNITS

Mixing kg/m³ with kN/m³.

γ = 9.81 × ρ / 1000.

USDA GRAVEL

Plotting raw percentages when gravel is present.

Re-scale by 100 − % gravel first.

THRESHOLDS

Mixing up AASHTO and USCS.

AASHTO 35% and No. 10. USCS 50% and No. 4.

ATTERBERG TESTS

Using the whole sample.

LL and PI are for the fraction passing No. 40.

EXAM DAY: 9 NOVEMBER

You’ve got this

Learn it, say it out loud, then test yourself. Repeat until the quick checks feel easy.

Do every quick check aloud

Redo the worked examples